RV College of Engineering®

Academic year 2025-2026 (Odd Sem)

DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING

(Common to CS, IS & AI)

Date: 10/11/2025	CIE - 1	Max. Marks: 60
Semester: V	UG	Duration: 2 Hour
Course Title: ADVANCED ALGORITHMS		Course Code: CS355TBB

Sl. No.	PART A	M	BT	CO
1.	Ω-notation represents the _____ bound of an algorithm's running time.	1	1	1
2.	The time complexity for $f(n) = 3n^2 + 5n + 1$ can be _____	2	2	1
3.	The recurrence relation $T(n)=T(n/2)+O(1)$ solves to _____	2	3	1
4.	The average time complexity of bucket sort can be given as _____	1	1	1
5.	If a dynamic array doubles its size when full, the amortized cost per insertion is_____	1	2	2
6.	_____ method in amortized analysis, assigns credits to operations	1	1	3
7.	The lower bound for comparison-based sorting algorithms can be given as _____	1	2	2
8.	Which function (n^3, n log n, 2^n, n^2) grows fastest?	1	2	3
	PART B			
1a.	Analyze how the Recursion Tree Method helps in determining the total work done in divide-and-conquer algorithms. Illustrate with an example such as $T(n)=2T(n/2)+ n\log n$.	6	3	1
1b.	The growth rates of some functions are: $f1(n) = n\log n$, $f2(n)=n^4$, $f3(n)=2n$, $f4(n)=n^2$ Analyze and arrange these functions in increasing order of growth rate. Justify using asymptotic notations.	4	3	2
2.	Apply Counting Sort algorithm fpr sorting the following list of integers: [4,2,2,8,3,3,1]. Show all intermediate steps including counting array and cumulative count. Write the algorithm and give the analysis.	10	3	4
3.	A dynamic table doubles its size when full and halves its size when one-fourth full. Using Accounting method of Amortized Analysis, compute the amortized cost of insertion and deletion operations.	10	2	2
4a.	Apply the Master Theorem to determine the asymptotic running time of the following recurrences: a)$T(n)=2T(n/2)+n$ b) $T(n)=3T(n/4)+n2$	6	3	1
4b.	"All comparison-based sorting algorithms require $\Omega(n \log n)$ comparisons in the worst case." Justify this statement.	4	2	3
5.	Apply Radix Sort algorithm to sort the following list of numbers in base 10 using Least Significant Digit (LSD) method: [329,457,657,839,436,720,355] Show the sorting after each digit pass. Write the algorithm and give the analysis.	10	3	1